# MATH 101 Unit 2: Functions and Their Graphs - Practice 2

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Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

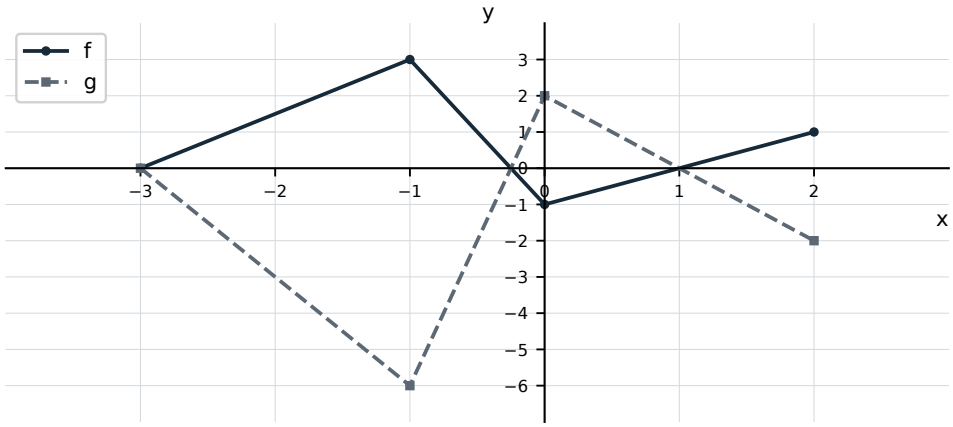
**1.** The solid graph is f and the dashed graph is g. Select the expression for g(x).

*A*. $g(x) = f(x) - 2$

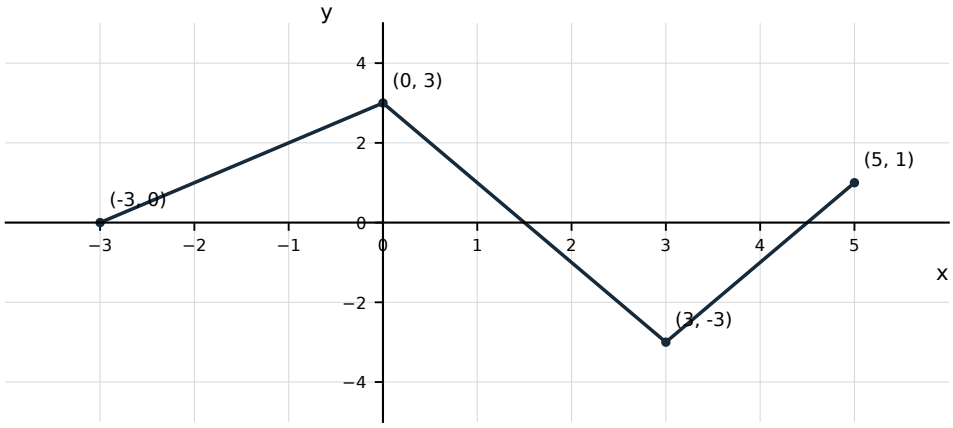
*B*. $g(x) = -2f(x)$

*C*. $g(x) = 2f(x)$

*D*. $g(x) = f(-2x)$




**2.** From the graph, find (a) average rate from x = -3 to x = 5; (b) increasing and decreasing intervals; (c) the two local extrema (ordered pairs).




**3.** (a) Find the linear function through these points, with domain all real numbers. (b) Find its inverse and give the inverse domain and range.

(1, 3); (3, 9)

**4.** For this function, find (a) f(-2), (b) domain, and (c) range.

$f(x) = 3|x - 2| - 3$

**5.** (a) Choose the pair giving h(x) = f(g(x)). (b) Form F(G(x)) and evaluate it at the supplied input.

$h(x) = (3x + 2)^2 + 4$

$A.\quad f(x) = x^2 + 4,\ g(x) = 3x + 2$

$B.\quad f(x) = 3x + 2,\ g(x) = x^2 + 4$

$C.\quad f(x) = x^2 + 2,\ g(x) = 3x + 4$

$D.\quad f(x) = 3x^2 + 4,\ g(x) = x + 2$

$(b)\quad F(x) = x^2,\ G(x) = x - 3,\ x_0 = 4$

# Student answer sheet

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**Question 1**

Choice

20 pts

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**Question 2**

(a) Rate

6 pts

(b) Increasing

4 pts

(b) Decreasing

4 pts

(c) Local max

3 pts

(c) Local min

3 pts

# Student answer sheet

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**Question 3**

(a) Function

10 pts

(b) Inverse

6 pts

(b) Domain

2 pts

(b) Range

2 pts

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**Question 4**

(a) Value

6 pts

(b) Domain

7 pts

(c) Range

7 pts

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**Question 5**

(a) Choice

8 pts

(b) Composition

8 pts

(b) Value

4 pts

# Answer key

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**Question 1**

| | | |
|---|---|---|
| Choice | B | 20 pts |

**Question 2**

| | | |
|---|---|---|
| (a) Rate | $\frac{1}{8}$ | 6 pts |
| (b) Increasing | $(-3, 0) \cup (3, 5)$ | 4 pts |
| (b) Decreasing | $(0, 3)$ | 4 pts |
| (c) Local max | $(0,\ 3)$ | 3 pts |
| (c) Local min | $(3,\ -3)$ | 3 pts |

# Answer key

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**Question 3**

| | | |
|---|---|---|
| (a) Function | $f(x) = 3x$ | 10 pts |
| (b) Inverse | $f^{-1}(x) = \frac{x}{3}$ | 6 pts |
| (b) Domain | $(-\infty, \infty)$ | 2 pts |
| (b) Range | $(-\infty, \infty)$ | 2 pts |

**Question 4**

| | | |
|---|---|---|
| (a) Value | $9$ | 6 pts |
| (b) Domain | $(-\infty, \infty)$ | 7 pts |
| (c) Range | $[-3, \infty)$ | 7 pts |

**Question 5**

| | | |
|---|---|---|
| (a) Choice | A | 8 pts |
| (b) Composition | $x^2 - 6x + 9$ | 8 pts |
| (b) Value | $1$ | 4 pts |